

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Mini-Project Synopsis

Department: - Computer Science and Engineering

Course Name: - Mini Project – Research Based / Mini Project

Course Code: - B22CI0605 / B22EI0605

**Title of the Project: - FitLens AI: Efficient Body Measurement Using
Camera Vision for Personalized Apparel, Fitness, and E-Commerce**

Group No. 3

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1. Problem Statement

Obtaining accurate body measurements is essential for applications such as clothing size selection and custom garment design. However, traditional methods of measuring the human body using measuring tapes are often inconvenient, time-consuming, and prone to human error. Many individuals struggle to measure their body dimensions correctly without assistance, leading to inaccurate measurements. This creates challenges in achieving proper clothing fit, especially in online shopping. Therefore, the difficulty in obtaining reliable and accurate body measurements remains a significant problem.

2. Methodology

1. Image Acquisition

The first step in the methodology is collecting user images.

- The user uploads two full-body photographs:
 - Front view
 - Side view
- Images are captured using a smartphone camera or webcam.
- The user can also use live camera for capturing photo.
- The user also provides their actual height in the image for calibration.
- The system validates that the images are clear, full-body, and properly aligned before processing.

2. Image Preprocessing

After image upload, preprocessing is performed to improve the quality of the input data.

- **Background removal and body segmentation** using YOLO.
- **Noise reduction and normalization** using OpenCV.
- Image resizing and contrast adjustments for better detection.
- Cropping to focus only on the human body region.

This step ensures that the AI model works with **clean and standardized images** for accurate detection.

3. Body Landmark Detection

The system then detects key body landmarks using **MediaPipe Pose**.

- MediaPipe identifies important **human body key points** such as:
 - Shoulders
 - Hips
 - Waist
 - Knees
 - Ankles
- These landmarks form a **skeletal representation of the body**, allowing the system to understand body proportions.

Landmark detection is crucial because all measurements are calculated based on the **distances between these key points**.

4. Pixel-to-Scale Conversion

Measurements extracted from images are initially in **pixel units**, which must be converted to real-world units.

- The system calculates a **pixel-to-centimeter ratio** using the user's **actual height**
- This scale factor allows conversion of **pixel distances to real-world measurements**.

5. Measurement Calculation

Once scaling is completed, the system calculates body measurements using geometric relationships between detected landmarks.

Key measurements include:

- Height
- Chest circumference
- Waist circumference
- Hip circumference

- Shoulder width
- Inseam length

Distances between landmarks are computed using **OpenCV mathematical operations** and converted into **centimeters and inches**.

6. Result Generation and Visualization

After calculations, the system generates the measurement results.

- Measurements are displayed on the **web interface**.
- The user can:
 - View measurements instantly.
 - Download a **measurement report**.
 - Use the data for **online shopping, tailoring, or fitness tracking**.

The system aims to provide **approximately 90–95% measurement accuracy**.

7. Data Privacy and Security

To ensure user trust and compliance with privacy standards:

- User photos are processed **only after consent**.
- Images are **encrypted during upload**.
- Data is **automatically deleted after processing**.
- No images are stored or reused without permission.

8. System Deployment

The system is deployed using a **web-based architecture**.

- **Frontend:** Browser-based interface for image upload and result display.
- **Backend:** AI models and processing pipeline.
- **Processing:** Cloud-based server for running computer vision algorithms.
- This architecture allows the system to **handle large numbers of users and scale easily**.

3. Software / Hardware Requirements

Hardware Requirements

- Minimum 4 GB RAM
- Recommended 8 GB RAM
- At least 2 GB of free disk space
- Webcam or mobile camera

Software Requirements

- Windows 10 or higher
- Python 3.9 – 3.12
- Node.js
- Git
- Visual Studio Code

4. Deliverables

- A web-based application that estimates human body measurements using AI and computer vision techniques.
- Implementation of body landmark detection using a pose estimation model to identify key body points.
- A measurement estimation module that calculates body dimensions such as height, chest, waist, and hips from images.
- A user interface that allows users to upload images and view their estimated body measurements.
- A final project report documenting the system design, methodology, and results.

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